

A TIME-AVERAGED BERNOULLI IDENTITY FOR TYPE I PROFILES OF THE NAVIER-STOKES EQUATIONS

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ABSTRACT. In backward similarity variables, the head pressure $\Pi = P + \frac{1}{2}|U|^2 + \frac{1}{2}y \cdot U$ of a Navier–Stokes profile satisfies $(\partial_s + L)\Pi + |\Omega|^2 = \partial_s P$, with $L = -\Delta + (U + \frac{1}{2}y) \cdot \nabla$. Pairing this identity with a time-dependent weight that solves the adjoint equation $\partial_s w = L^* w$ (a time-reversed Fokker–Planck equation, whose pullback invariant density we construct with two-sided Gaussian bounds) gives, for every profile with a global Type I bound and every time window,

$$\left\langle \int |\Omega|^2 w \, dy \right\rangle = - \left\langle \int P \partial_s w \, dy \right\rangle + O(1/T),$$

exactly over a period for discretely self-similar profiles. For stationary profiles the right side vanishes (Nečas–Růžička–Šverák, Tsai); for rotated profiles it is Pineau–Vicol’s small-rotation term. In general the whole obstruction to a Liouville theorem is a single averaged correlation between the pressure and the motion of the weight: if it is small, the profile is trivial, with no assumption on the period. A localized version, with a quasi-stationary weight on an expanding ball, gives a necessary condition for any Type I singularity in the sense of Pineau–Vicol: the localized correlation stays bounded below on average over every long window. Model computations show the correlation is small for thin-cored vortex clusters but grows with core size and amplitude without visible saturation, so it must be controlled by its sign or dynamics rather than its size.

1. INTRODUCTION

A backward self-similar solution $u(x, t) = (-t)^{-1/2}U(x/\sqrt{-t})$ of the three-dimensional Navier–Stokes equations would blow up at $t = 0$. Nečas, Růžička and Šverák [5] showed that no such profile exists in L^3 , using a maximum principle for the head pressure (Bernoulli function) $\Pi = P + \frac{1}{2}|U|^2 + \frac{1}{2}y \cdot U$; Tsai [7] extended this to profiles satisfying local energy estimates. Discretely self-similar profiles, periodic in the similarity time $s = -\log(-t)$, are excluded when the period is short (Chae–Wolf [2]). Recently Pineau and Vicol [6] replaced the maximum principle by a weighted L^2 identity: the adjoint of $L = -\Delta + (U + \frac{1}{2}y) \cdot \nabla$ has a positive kernel element w with Gaussian bounds, and pairing $L\Pi + |\Omega|^2 = E$ with w gives $\int |\Omega|^2 w = \int Ew$. This proves triviality for rotated self-similar profiles with small or large rotation, for discretely self-similar profiles with factor close to 1, and a regularity criterion for Type I solutions that are approximately self-similar at one time.

For time-dependent profiles, [6] pair the identity with a weight built from the time-averaged drift and control the fluctuations by the length of the period. Here we use a weight that moves with the profile: a solution of $\partial_s w = L^*(s)w$. This removes the time derivative of the head pressure exactly, up to a boundary term in time that averages out, and yields an identity in which the only remaining term is the correlation $\langle \int P \partial_s w \rangle$ (Theorem 4.1). The weight is the pullback invariant density of an Ornstein–Uhlenbeck process perturbed by $-U$, and is periodic when U is (Lemma 3.1).

Results. (i) The time-dependent Bernoulli identity (Proposition 2.1). (ii) The eternal adjoint weight, with two-sided Gaussian bounds (Lemma 3.1). (iii) The averaged weighted-entropy identity for every profile with a global Type I bound (Theorem 4.1). (iv) A Liouville criterion: if the averaged correlation is below an explicit $\delta_0(C_u)$, the profile is trivial; for discretely self-similar profiles this holds for every factor $\lambda > 1$, not only λ near 1 (Corollary 4.2). (v) A localized version for Type I singularities (Theorem 5.2). (vi) Size bounds showing that the correlation cannot be controlled by size alone, and model computations supporting this (Section 7).

Prior art. Time-periodic invariant measures of time-periodic diffusions are classical (Khasminskii [4]). We found no prior use of a forward-in-time adjoint weight for backward Navier–Stokes profiles; [6] use a fixed averaged weight (§7 there) and a single-time eigenfunction (§9 there).

2. SETTING AND THE BERNOULLI IDENTITY

Let u solve the Navier–Stokes equations ($\nu = 1$) on $\mathbb{R}^3 \times [-1, 0)$ and set $y = x/\sqrt{-t}$, $s = -\log(-t)$, $U(y, s) = \sqrt{-t}u$, $P(y, s) = (-t)p$. Then

$$\partial_s U + \frac{1}{2}U + \frac{1}{2}(y \cdot \nabla)U + (U \cdot \nabla)U + \nabla P - \Delta U = 0, \quad \nabla \cdot U = 0. \quad (1)$$

The *global Type I bound* is $|u(x, t)| \leq C_u/(\sqrt{-t} + |x|)$ on $\mathbb{R}^3 \times [-1, 0)$, i.e. $|U(y, s)| \leq C_u/(1 + |y|)$; it implies $|\nabla^k U| \leq C_{U,k}/(1 + |y|)^{k+1}$, $|\partial_s U| \leq C_{U,s}$, and $|P| \leq C_{P,0}$ ([6], (2.1)–(2.2), (7.1)–(7.3)). Write $\tilde{U} = U + \frac{1}{2}y$, $\Omega = \nabla \times U$, $\Pi = P + \frac{1}{2}|U|^2 + \frac{1}{2}y \cdot U$, $L = -\Delta + \tilde{U} \cdot \nabla$ and $L^*f = -\Delta f - \tilde{U} \cdot \nabla f - \frac{3}{2}f$.

Proposition 2.1 (time-dependent Bernoulli identity). $(\partial_s + L)\Pi + |\Omega|^2 = \partial_s P$.

Proof. Since $(U \cdot \nabla)y = U$ and $(y \cdot \nabla)y = y$, $(\tilde{U} \cdot \nabla)\tilde{U} = (U \cdot \nabla)U + \frac{1}{2}(y \cdot \nabla)U + \frac{1}{2}U + \frac{1}{4}y$, so (1) reads $\partial_s U + (\tilde{U} \cdot \nabla)\tilde{U} - \nabla \frac{|y|^2}{8} + \nabla P - \Delta U = 0$. With $(\tilde{U} \cdot \nabla)\tilde{U} = \nabla \frac{1}{2}|\tilde{U}|^2 - \tilde{U} \times \Omega$ ($\nabla \times \tilde{U} = \Omega$), $\frac{1}{2}|\tilde{U}|^2 - \frac{1}{8}|y|^2 + P = \Pi$ and $-\Delta U = \nabla \times \Omega$,

$$\nabla \Pi = \tilde{U} \times \Omega - \nabla \times \Omega - \partial_s U.$$

The divergence gives $\Delta \Pi = |\Omega|^2 - \tilde{U} \cdot \nabla \times \Omega$, and the dot product with $\tilde{U}$ gives $\tilde{U} \cdot \nabla \Pi = -\tilde{U} \cdot \nabla \times \Omega - \tilde{U} \cdot \partial_s U$. Subtracting, $L\Pi + |\Omega|^2 = -\tilde{U} \cdot \partial_s U = -\partial_s(\frac{1}{2}|U|^2 + \frac{1}{2}y \cdot U) = -\partial_s(\Pi - P)$. $\square$

For stationary profiles this is $L\Pi = -|\Omega|^2$, the identity of [5, 7, 6]; [6], (7.7)–(7.8), is the same identity written for rotated discretely self-similar profiles.

3. THE ETERNAL ADJOINT WEIGHT

Lemma 3.1. Assume the global Type I bound. There is a positive solution w of $\partial_s w = L^*(s)w$ on $\mathbb{R}^3 \times \mathbb{R}$ with $\int w(\cdot, s) dy = 1$, which for every $\delta \in (0, \frac{1}{2}]$ satisfies

$$m e^{-(1+\delta)|y|^2/4} \leq w(y, s) \leq M e^{-(1-\delta)|y|^2/4}, \quad |\nabla w| + |\nabla^2 w| + |\partial_s w| \leq M' e^{-(1-2\delta)|y|^2/4},$$

with m, M, M' depending only on δ and the constants of the Type I bound. It is the unique such solution, and it is periodic when U is periodic in s .

Proof. Write $\sigma = -s$, $\rho(\cdot, \sigma) = w(\cdot, -\sigma)$. Then $\partial_s w = L^*w$ becomes $\partial_\sigma \rho = \Delta \rho + \nabla \cdot (\tilde{U} \rho)$, the Fokker–Planck equation of $dX = -\tilde{U} d\sigma + \sqrt{2} dB$: an Ornstein–Uhlenbeck process with an added drift $-U$, smooth, divergence-free, bounded by $C_u/(1 + |y|)$.

Moments. For a probability solution, $M_2 = \int |y|^2 \rho$ satisfies $\dot{M}_2 = \int \rho(6 - 2y \cdot U - |y|^2) \leq 6 + 2C_u - M_2$. For $0 < a < \frac{1}{4}$ and $V_a = e^{a|y|^2}$, $\Delta V_a - \tilde{U} \cdot \nabla V_a = aV_a(6 + (4a - 1)|y|^2 - 2y \cdot U) \leq aV_a(6 + 2C_u - (1 - 4a)|y|^2)$, so $\frac{d}{d\sigma} \int V_a \rho \leq C_a - c_a \int V_a \rho$, uniformly in time.

Pullback limit. Let ρ_{σ_0} start from the Gaussian $G = (4\pi)^{-3/2}e^{-|y|^2/4}$ at σ_0 . V_a is a uniform Lyapunov function, and on each ball the unit-time transition densities are bounded below by the parabolic Harnack inequality. The time-inhomogeneous Harris theorem [3] gives contraction in the V_a -weighted total-variation norm, $\|\rho_{\sigma_0}(\sigma) - \rho_{\sigma_1}(\sigma)\|_{V_a} \leq C\theta^{\sigma - \max(\sigma_0, \sigma_1)}$, $\theta < 1$; so $\rho_{\sigma_0}(\sigma)$ converges as $\sigma_0 \rightarrow -\infty$ to a probability solution ρ on $\mathbb{R}^3 \times \mathbb{R}$, unique among solutions bounded in this norm and hence periodic for periodic drift. It is smooth by parabolic regularity.

Gaussian bounds. For $\phi_\beta = e^{-\beta|y|^2}$, using $\nabla \cdot \tilde{U} = \frac{3}{2}$,

$$\Delta\phi_\beta + \nabla \cdot (\tilde{U}\phi_\beta) = \phi_\beta(\beta(4\beta - 1)|y|^2 + \frac{3}{2} - 6\beta - 2\beta y \cdot U).$$

For $\beta = \frac{1+\delta}{4}$ the bracket is $\geq \frac{\delta(1+\delta)}{4}|y|^2 - C$, so ϕ_β is a stationary subsolution outside a ball B_{R_δ} ; for $\beta = \frac{1-\delta}{4}$ it is a supersolution there. The moment bound gives $\int_{B_R} \rho \geq \frac{1}{2}$ for some $R = R(C_u)$, and the Harnack inequality gives $\rho \geq c$ on $B_{R_\delta} \times \mathbb{R}$. Comparison on $\{|y| > R_\delta\} \times [\sigma_0, \sigma]$, with these boundary values, and $\sigma_0 \rightarrow -\infty$ (the initial data are forgotten by the contraction) give the lower bound; the upper bound follows in the same way from the supersolution, local bounds on ∂B_{R_δ} , and $G \leq C\phi_{(1-\delta)/4}$. The derivative bounds follow from local parabolic estimates on balls of radius $(1 + |y|)^{-1}$, where the drift is of size $1 + |y|$. $\square$

4. THE AVERAGED IDENTITY AND A LIOUVILLE CRITERION

Theorem 4.1. *Assume the global Type I bound and let w be the weight of Lemma 3.1. For every $s_0 \geq 1$ and $T > 0$,*

$$\frac{1}{T} \int_{s_0}^{s_0+T} \int_{\mathbb{R}^3} |\Omega|^2 w \, dy \, ds = -\frac{1}{T} \int_{s_0}^{s_0+T} \int_{\mathbb{R}^3} P \partial_s w \, dy \, ds + \frac{\mathcal{B}(s_0) - \mathcal{B}(s_0 + T)}{T},$$

with $\mathcal{B}(s) = \int (\Pi - P)w \, dy$, $|\mathcal{B}| \leq C_u + \frac{1}{2}C_u^2$. For a periodic profile and T a multiple of the period the last term vanishes.

Proof. Pair Proposition 2.1 with w ; boundary terms at infinity vanish by the Gaussian decay of w and the bounds on Π , P . Since $\partial_s w = L^* w$, $\frac{d}{ds} \int \Pi w = \int (\partial_s \Pi + L\Pi)w = \int (\partial_s P - |\Omega|^2)w = \frac{d}{ds} \int Pw - \int P \partial_s w - \int |\Omega|^2 w$, i.e. $\frac{d}{ds} \mathcal{B} + \int |\Omega|^2 w = -\int P \partial_s w$. Integrate in s ; $|\Pi - P| = \frac{1}{2}|U|^2 + \frac{1}{2}y \cdot U \leq \frac{1}{2}C_u^2 + \frac{1}{2}C_u$ and $\int w = 1$. $\square$

Corollary 4.2. *Let $\eta \in (0, 1]$ and R_η, s_R be the constants of [6], Lemma 9.4, and set $\delta_0 = \frac{\eta^2}{8} m e^{-(1+\delta)R_\eta^2}$ with $m = m(\delta, C_u)$ from Lemma 3.1 ($\delta = \frac{1}{2}$). Assume the global Type I bound. If for some $s_0 \geq s_{R_\eta}$ and $T \geq 4(C_u + C_u^2)/\delta_0$,*

$$\frac{1}{T} \left| \int_{s_0}^{s_0+T} \int P \partial_s w \, dy \, ds \right| \leq \frac{\delta_0}{2},$$

then $(0, 0)$ is a regular point. In particular a discretely self-similar solution with any factor $\lambda > 1$, or a rotated self-similar solution with any rotation, satisfying this condition over one period, is identically zero.

Proof. By Theorem 4.1 the average of $\int |\Omega|^2 w$ over $[s_0, s_0 + T]$ is at most δ_0 , so some s_* in the window has $\int |\Omega(s_*)|^2 w(s_*) \leq \delta_0$. Since $w \geq m e^{-(1+\delta)R_\eta^2}$ on B_{2R_η} , $\|\Omega(s_*)\|_{L^2(B_{2R_\eta})}^2 \leq \eta^2/8 < (\eta/2)^2$. By [6], Lemma 9.4 and Proposition 9.5 (propagation of small vorticity and the Caffarelli–Kohn–Nirenberg criterion [1]), $(0, 0)$ is regular. A discretely self-similar solution bounded near $(0, 0)$ satisfies $u(x, t) = \lambda^k u(\lambda^k x, \lambda^{2k} t) \rightarrow 0$ as $k \rightarrow \infty$; rotated self-similar solutions are discretely self-similar ([6], Remark 1.5). $\square$

Remark 4.3. For stationary profiles $\partial_s w = 0$ and Corollary 4.2 is Tsai's theorem; for rotated profiles with rotation α , $\partial_s w = -\alpha(Jy \cdot \nabla)w$ in the rotating frame and $\int P \partial_s w = -\alpha \int w \partial_\theta P$, which recovers the small- α case of [6]. Since the axisymmetric part of w does not contribute, only the non-axisymmetric part of the weight enters.

5. A LOCALIZED VERSION

Let (u, p) be smooth on $B_1 \times [-1, 0)$ with the local Type I bound $|u| \leq C_u/(\sqrt{-t} + |x|)$ and p bounded on $A \times [-1, 0)$, $A = \{\frac{1}{2} < |x| < \frac{3}{4}\}$ (the hypotheses of [6], Theorem 1.9). The profile is defined on $\{|y| < e^{s/2}\}$; set $R(s) = \frac{5}{8}e^{s/2}$. By [6], (9.17)–(9.19), for $s \geq s_0(C_u, C_p)$,

$$|\Pi| \leq 2C_u \text{ on } \partial B_{R(s)}, \quad \|P(\cdot, s)\|_{L^6(B_{R(s)})} \leq C'_P.$$

Lemma 5.1 (localized weight). *Let $s_* \leq s_a < s_b$, $T = s_b - s_a$, $D(s) = B_{R(s)}$. Let v solve, backwards in s ,*

$$\partial_s v = L^* v \text{ in } D(s), \quad v = 0 \text{ on } \partial D(s), \quad v(\cdot, s_b) = v_b := (e^{-|y|^2/4} - e^{-R(s_b)^2/4})_+,$$

and set $m(s) = \int_{D(s)} v$, $w = v/m$. Then:

- (1) $\partial_s w = L^* w - \gamma w$ in $D(s)$, $w = 0$ on $\partial D(s)$, $\int_{D(s)} w = 1$, with $\gamma(s) = \oint_{\partial D(s)} |\partial_n w| \geq 0$; and for $f \in C^2(\overline{D(s)})$,

$$\frac{d}{ds} \int_{D(s)} f w = \int_{D(s)} (\partial_s f + Lf) w - \gamma \int_{D(s)} f w + \oint_{\partial D(s)} f |\partial_n w|. \quad (2)$$

- (2) (exit rate) There are $c, C > 0$ depending on C_u such that $\gamma(s) \leq C e^{-cR(s)^2}$ for $s \in [s_a, s_b]$, provided $s_* \geq s_*(T, C_u)$ (a bound of the form $2 \log \log(CT) + C$).
- (3) (Gaussian bounds) For $\delta \in (0, \frac{1}{2}]$ there are $m_\delta, M_\delta > 0$ depending only on δ, C_u such that $w \leq M_\delta e^{-(1-\delta)|y|^2/4}$ on $D(s)$ and $w \geq m_\delta e^{-(1+\delta)|y|^2/4}$ on $B_{R(s)/2}$, for all $s \in [s_a, s_b]$.
- (4) (derivatives) $|\nabla w| + |\partial_s w| \leq C_\delta (1 + |y|)^3 e^{-(1-\delta)|y|^2/4}$ on $D(s)$.

Proof. Throughout, $\tilde{U} = U + \frac{1}{2}y$ with $|U| \leq C_u/(1 + |y|)$ and $|\nabla^k U| \leq C_{U,k}$ on $\{|y| \leq \frac{3}{4}e^{s/2}\} \supset \overline{D(s)}$ ([6], (9.4)); $\nabla \cdot \tilde{U} = \frac{3}{2}$, and $L^* v = -\Delta v - \nabla \cdot (\tilde{U} v)$. In the forward time $\theta = s_b - s$ the equation is the Fokker–Planck equation $\partial_\theta v = \Delta v + \nabla \cdot (\tilde{U} v)$ of $dX = -\tilde{U} d\theta + \sqrt{2} dB$, killed on the boundary of the shrinking domain $D(s_b - \theta)$.

Step 0: existence and positivity. With $z = y/R(s)$ the domain becomes the unit ball and the equation becomes a uniformly parabolic equation in z with smooth bounded coefficients on $[s_a, s_b]$ (the moving boundary contributes the smooth drift $-\frac{1}{2}z \cdot \nabla_z$); classical theory gives a unique solution, smooth up to the boundary for $\theta > 0$. Since $v_b \geq 0$, $v_b \not\equiv 0$, the strong maximum principle gives $v > 0$ in $D(s)$ for $s < s_b$, and the Hopf lemma gives $\partial_n v < 0$ on $\partial D(s)$.

Step 1: normalization and the pairing identity. Since $v = 0$ on the moving boundary, $\frac{d}{ds} \int_{D(s)} g = \int_{D(s)} \partial_s g$ for any g vanishing on $\partial D(s)$. Hence $\dot{m} = \int L^* v = -\oint \partial_n v - \oint (\tilde{U} \cdot n)v = \oint |\partial_n v|$, and $w = v/m$ satisfies $\partial_s w = L^* w - \gamma w$ with $\gamma = \dot{m}/m = \oint |\partial_n w|$. For (2): $\frac{d}{ds} \int f w = \int (\partial_s f) w + \int f (L^* w - \gamma w)$, and two integrations by parts give $\int f (-\Delta w) = \int (-\Delta f) w - \oint f \partial_n w$ and $\int f (-\nabla \cdot (\tilde{U} w)) = \int (\tilde{U} \cdot \nabla f) w$ (both boundary terms with a factor w vanish), so $\int f L^* w = \int (L f) w + \oint f |\partial_n w|$.

Step 2: upper bound away from the boundary layer. Fix $\beta = \frac{1-\delta}{4}$ and $\phi_\beta = e^{-\beta|y|^2}$. As in Lemma 3.1, $\Delta \phi_\beta + \nabla \cdot (\tilde{U} \phi_\beta) = \phi_\beta (-\beta \delta |y|^2 + \frac{3}{2} - 6\beta - 2\beta y \cdot U) < 0$ for $|y| \geq R_\delta = R_\delta(C_u)$: a stationary supersolution of the forward equation there. Mass is non-increasing in θ ($\frac{dm}{d\theta} = -\oint |\partial_n v| \leq 0$), so $m(s) \leq m_b := m(s_b)$. Interior parabolic estimates on B_{2R_δ}

(coefficients bounded there) give $\sup_{B_{R_\delta}} v(\cdot, s) \leq C_\delta m_b$ for $s \leq s_b - 1$, and directly from v_b for $s \in [s_b - 1, s_b]$ (comparison with the bound $e^{(\frac{3}{2} + C_u)\theta} e^{-\beta|y|^2}$, valid on D for $\theta \leq 1$). Comparison on $\{R_\delta < |y| < R(s)\}$, where $v \leq C_\delta m_b e^{\beta R_\delta^2} \phi_\beta$ on the inner boundary, $v = 0 < \phi_\beta$ on the outer one, and $v_b \leq \phi_\beta$ initially, gives

$$v \leq C'_\delta m_b e^{-(1-\delta)|y|^2/4} \quad \text{on } D(s), \quad s \in [s_a, s_b].$$

Step 3: exit rate and mass. In the boundary layer $\{R(s) - 1 < |y| < R(s)\}$ the drift has size $\lesssim R$; rescaling a unit neighbourhood of a boundary point by R^{-1} and applying boundary gradient estimates for the Dirichlet problem gives $|\partial_n v| \leq CR^2 \sup_{|y| > R-1} v \leq CR^2 m_b e^{-(1-\delta)(R-1)^2/4}$ on $\partial D(s)$. Hence $\oint |\partial_n v| \leq CR^4 m_b e^{-(1-\delta)(R-1)^2/4}$, and integrating $\dot{m} = \oint |\partial_n v|$ over $[s, s_b]$, $m(s) \geq m_b (1 - CT R(s_a)^4 e^{-(1-\delta)(R(s_a)-1)^2/4}) \geq \frac{1}{2} m_b$ once $s_a \geq s_*(T, C_u)$, which needs only $R(s_a)^2 \gtrsim \log(CT)$. Then $\gamma = \dot{m}/m \leq 2CR^4 e^{-(1-\delta)(R-1)^2/4} \leq C e^{-cR^2}$, which is (2); and $w = v/m \leq 2C'_\delta e^{-(1-\delta)|y|^2/4}$, the upper bound in (3).

Step 4: interior lower bound. The second moment $N(s) = \int |y|^2 w$ satisfies $\frac{dN}{d\theta} = \int w(6 - 2y \cdot U - |y|^2) - \oint |y|^2 |\partial_n w| + \gamma N \leq 6 + 2C_u - N + \gamma N$, so $N \leq \max(N(s_b), 7 + 2C_u)$ for $\gamma \leq \frac{1}{2}$; hence $\int_{B_{R_1}} w \geq \frac{1}{2}$ with $R_1 = R_1(C_u)$, for all s . The parabolic Harnack inequality on $B_{2R_1+2R_\delta}$ then gives $w \geq c_\delta > 0$ on B_{R_δ} for $s \leq s_b - 1$, and the same holds for $s \in [s_b - 1, s_b]$ from the explicit v_b and short-time continuity. For the outer region take $\beta' = \frac{1+\delta}{4}$, $k(s) = e^{-\beta' R(s)^2}$ and $\psi = \phi_{\beta'} - k$. Then $\psi = 0$ on $\partial D(s)$, and in the forward time (k increases with θ , since R decreases)

$$\Delta \psi + \nabla \cdot (\tilde{U} \psi) - \partial_\theta \psi \geq \phi_{\beta'} (\beta' \delta |y|^2 - C) - \frac{3}{2} k \geq 0 \quad \text{on } \{R_\delta \leq |y| < R(s)\},$$

because $\phi_{\beta'} \geq k$ there and $\beta' \delta |y|^2 - C \geq \frac{3}{2}$ for $|y| \geq R_\delta$ (enlarging R_δ). So ψ is a subsolution on the annulus; it is $\leq e^{-\beta' R_\delta^2} \leq c_\delta^{-1} e^{-\beta' R_\delta^2} w$ on the inner boundary, $0 = w$ on the outer one, and $\leq C v_b / m_b$ initially. Comparison gives $w \geq c'_\delta (\phi_{\beta'} - k) \geq \frac{1}{2} c'_\delta e^{-(1+\delta)|y|^2/4}$ on $B_{R(s)/2}$, since there $k \leq e^{-3\beta' R^2/4} \phi_{\beta'} \leq \frac{1}{2} \phi_{\beta'}$. This is the lower bound in (3).

Step 5: derivatives. Interior parabolic estimates on balls of radius $(1 + |y|)^{-1}$ (where the drift is of size $1 + |y|$), applied to w with the bound of Step 3, give $|\nabla w| + |\nabla^2 w| \leq C(1 + |y|)^2 e^{-(1-\delta)|y|^2/4}$ away from the boundary layer, and the boundary estimates of Step 3 cover the layer, where w is exponentially small in R^2 . Then $|\partial_s w| = |L^* w - \gamma w| \leq C(1 + |y|)^3 e^{-(1-\delta)|y|^2/4}$. $\square$

Theorem 5.2 (a necessary condition for Type I blowup). *Under the hypotheses above there are $\delta_0 > 0$, s_* and T_0 , depending only on C_u and C_p , such that, with w the localized weight of Lemma 5.1 on $[s_a, s_a + T]$: if for some $T \geq T_0$ and $s_a \geq s_*(T)$*

$$\frac{1}{T} \left| \int_{s_a}^{s_a+T} \int_{B_{R(s)}} P \partial_s w \, dy \, ds \right| \leq \frac{\delta_0}{2},$$

then $(0, 0)$ is a regular point. Equivalently, at a Type I singularity the localized pressure-weight correlation averages to more than $\delta_0/2$ over every window of length $T \geq T_0$ starting after s_ .*

Proof. Proposition 2.1 holds pointwise on $\{|y| < e^{s/2}\} \supset \overline{D(s)}$. Apply (2) with $f = \Pi$ and with $f = P$ (the latter in the form $\frac{d}{ds} \int P w = \int (\partial_s P) w + \int P \partial_s w$, valid since $w = 0$ on $\partial D(s)$):

$$\frac{d}{ds} \int (\Pi - P) w + \int |\Omega|^2 w = - \int P \partial_s w - \gamma \int \Pi w + \oint_{\partial D(s)} \Pi |\partial_n w|.$$

The last two terms are bounded by $\sup |\Pi| \gamma + 2C_u \gamma \leq C \gamma(s) \leq C e^{-cR(s)^2}$ (using $|\Pi| \leq C$ on the ball, from $|\Pi - P| \leq C$ and the L^6 bound on P paired with $w \in L^{6/5}$, and $|\Pi| \leq 2C_u$ on

the boundary); $\int |P \partial_s w| \leq C'_P \|\partial_s w\|_{L^{6/5}} < \infty$. Integrating over the window, the average of $\int |\Omega|^2 w$ is at most $\delta_0/2 + 2(C_u + C_u^2)/T + Ce^{-cR(s_a)^2} \leq \delta_0$ for $T \geq T_0$ and $s_a \geq s_*$. The rest is the proof of Corollary 4.2, with Proposition 9.5 of [6] applied to the local solution. $\square$

Remark 5.3. Theorem 5.2 and [6], Theorem 1.9, are complementary. Theirs needs the profile to be nearly stationary at a single time; ours needs the profile's motion to be nearly uncorrelated with the pressure on average, and allows $\partial_s U$ to be large. A Type I singularity must defeat both: its profile moves at a uniform rate forever, and that motion stays correlated with the pressure.

6. ROTATED PROFILES: A ROTATING-FRAME FORM

A rotated self-similar profile [6] is $U(y, s) = R(\alpha s)U^*(R(-\alpha s)y)$, $R(\theta)$ the rotation by θ about e_3 ; in the rotating frame U^* is stationary and solves $\alpha(JU - (Jy \cdot \nabla)U) + \frac{1}{2}U + \frac{1}{2}(y \cdot \nabla)U + (U \cdot \nabla)U + \nabla P - \Delta U = 0$ (dropping stars), with $Jy = e_3 \times y$. Set $W = U + \frac{1}{2}y - \alpha Jy$, $L_W = -\Delta + W \cdot \nabla$, and $\Pi_\alpha = \Pi - \alpha(Jy) \cdot U$.

Proposition 6.1 (rotating-frame Bernoulli identity). $L_W \Pi_\alpha + |\Omega|^2 = 2\alpha \Omega_3$, and Π_α is bounded.

Proof. A direct expansion gives $(W \cdot \nabla)W = (\tilde{U} \cdot \nabla)\tilde{U} - \alpha(Jy \cdot \nabla)U - \alpha JU - \alpha Jy - \alpha^2 y_\perp$, $y_\perp = (y_1, y_2, 0)$, so the profile equation becomes $(W \cdot \nabla)W + 2\alpha JW - \nabla(\frac{\alpha^2}{2}|y_\perp|^2 + \frac{1}{8}|y|^2) + \nabla P - \Delta U = 0$: a Coriolis term $2\alpha e_3 \times W$ and a centrifugal potential. Since $\nabla \times W = \Omega - 2\alpha e_3$, $(W \cdot \nabla)W + 2\alpha e_3 \times W = \nabla \frac{1}{2}|W|^2 - W \times \Omega$: the Coriolis term cancels the frame vorticity. With $\frac{1}{2}|W|^2 - \frac{\alpha^2}{2}|y_\perp|^2 - \frac{1}{8}|y|^2 + P = \Pi_\alpha$ (the quadratic terms cancel, leaving $\Pi_\alpha = P + \frac{1}{2}|U|^2 + \frac{1}{2}y \cdot U - \alpha Jy \cdot U$, bounded by the Type I bound) and $-\Delta U = \nabla \times \Omega$, we get $\nabla \Pi_\alpha = W \times \Omega - \nabla \times \Omega$. Then $\Delta \Pi_\alpha = \Omega \cdot (\Omega - 2\alpha e_3) - W \cdot \nabla \times \Omega$ and $W \cdot \nabla \Pi_\alpha = -W \cdot \nabla \times \Omega$; subtract. $\square$

Corollary 6.2. Let w_α be the normalized positive kernel element of $L_W^* f = -\Delta f - \nabla \cdot (Wf)$ (Lemma 3.1 with the divergence-free drift $-\alpha Jy$ added; the Gaussian is invariant under it). Then

$$\int_{\mathbb{R}^3} |\Omega - \alpha e_3|^2 w_\alpha dy = \alpha^2, \quad \text{equivalently} \quad \int |\Omega|^2 w_\alpha = 2\alpha \int \Omega_3 w_\alpha.$$

Hence $\int |\Omega|^2 w_\alpha \leq 4\alpha^2$, and with Corollary 4.2 (or [6], Proposition 3.1) the profile is trivial for $4\alpha^2 < \delta_0$, a threshold of order the square root of the one obtained from a bound linear in $|\alpha|$.

Proof. Pair Proposition 6.1 with w_α : $\int (L_W \Pi_\alpha) w_\alpha = \int \Pi_\alpha L_W^* w_\alpha = 0$. Then $\int |\Omega - \alpha e_3|^2 w = \int |\Omega|^2 w - 2\alpha \int \Omega_3 w + \alpha^2 \int w = \alpha^2$. Cauchy-Schwarz gives $\int |\Omega|^2 w \leq 2|\alpha|(\int |\Omega|^2 w)^{1/2}$. $\square$

Remark 6.3 (moderate rotation). With the rotating-frame energy identity $\int (\frac{1}{2}|U|^2 + |\nabla U|^2) w_\alpha = \int P U \cdot \nabla w_\alpha$ and virial identity $\int (|U|^2 - y \cdot \nabla P) w_\alpha = 0$ (derived as in the non-rotating case), one finds $2 \int \Omega_3 w_\alpha = - \int w_\alpha \partial_\theta P$, so Corollary 6.2 is Theorem 4.1 again; viewing the profile from a frame rotating at another rate gives the same weight, rotated. For moderate α the question is whether a nontrivial profile can have $\int |\Omega - \alpha e_3|^2 w_\alpha = \alpha^2$, the value of the trivial one. The rotating-frame Bernoulli function (Coriolis does no work) is classical; we found no previous use of it for this problem.

7. SIZE BOUNDS AND MODEL COMPUTATIONS

Proposition 7.1. Under the global Type I bound, $|\int P \partial_s w| \leq C(C_u)$ for all s , so both sides of Theorem 4.1 are bounded; and a vortex tube in the profile with circulation Γ and Gaussian core radius ρ has $\rho^2 \geq \Gamma/(\pi C_{U,1})$.

Proof. $|P| \leq C_{P,0}$ and $\int |\partial_s w| \leq C$ by Lemma 3.1. A Gaussian core has peak vorticity $\Gamma/(\pi\rho^2) \leq \sup |\Omega| \leq 2C_{U,1}$. $\square$

These bounds come from the Type I gradient bound alone, so a contradiction with Theorem 4.1 cannot come from the size of the correlation; it needs its sign or its time structure. Model computations confirm this. We took profiles U^* made of vortex rings with Gaussian cores (core size a , circulation Γ), rotating at rate α about e_3 , computed the rotating-frame weight w^* (the stationary density of $dX = -(U^* + \frac{y}{2} - \alpha Jy)dt + \sqrt{2}dB$) and the ratio $Q = |\alpha \int w^* \partial_\theta P^*| / \int |\Omega^*|^2 w^*$; an exact rotated solution has $Q = 1$. Over three geometries, $a \in [0.4, 2]$, $\Gamma \in [2, 256]$, $\alpha \in [0.25, 2]$, Q ranged from 10^{-4} to 0.13: it is small for thin cores and small amplitude, grows with core size, amplitude and rotation, levels off in core size near $a \approx 1.2$ – 1.6 , but keeps growing with amplitude (roughly as $\Gamma^{0.4}$ between $\Gamma = 64$ and 256 , reaching 0.13). The non-axisymmetric part of the weight does not track the pressure ($|\text{corr}(\tilde{w}, \tilde{P})| \leq 0.14$); the smallness comes from weak alignment of $\tilde{w}$ with $\partial_\theta P$. These are model fields, not solutions, and say nothing about whether $Q = 1$ is attainable; they indicate only that the correlation is not structurally small in the fat-core regime, where strong vortices in a Type I profile must live (Proposition 7.1).

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